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DEPARTMENT OF MECHANICAL ENGINEERING

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Instructor: Assoc. Prof. Dr. Murat Özdenefe
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Assessment: Project
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PROJECT NAME: Heatsink Design

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SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract	1	1, 3, 6	10		
Introduction	1	1, 3, 6	10		
Model & Calculations	1	1, 3, 6	40		
Results & Discussion	1	1, 3, 6	20		
Conclusion	1	1, 3, 6	10		
References & Appendix	1	1, 3, 6	10		
					Marks for report format are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Understand the basic mechanisms of heat transfer.	X					X	
2	Be able to obtain and solve DE of heat conduction.	X						
3	Comprehend steady conduction, solve steady conduction problems by resistance concept and analyze fins.	X					X	
4	Be able to solve transient heat transfer problems for lumped systems & systems with spatial effects.	X						
5	Understand the numerical solution of steady, 1D & 2D and transient, 1D & 2D heat conduction problems.	X						
6	Evaluate convection coefficient, heat transfer and associated temperatures for external flow.	X					X	
7	Calculate convection coefficient, heat transfer and associated temperatures for internal flow.	X						
8	Evaluate convection coefficient and heat transfer for free convection cases.	X						
9	Analyze & size HEXs by LMTD and ϵ -NTU method.	X						
	Weight of Student Outcomes	H					M	

Instructions:

- A. Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source. Please include the relevant standards (preferably in the Appendix).
- B. Submission:** Upload your report as a pdf file to the "Lab Work and Project" section in LMS within the designated time. The file uploaded should be named as [MENG345-Project-Student IDs-Student Names]. No late submissions will be accommodated.

ABSTRACT

This project involves the design and thermal evaluation of a heat sink intended to cool a CPU that dissipates 110 W of heat while operating within a surface temperature limit of 60 °C. The heat sink must fit a base area of 80 mm × 80 mm and a maximum height of 35 mm. Key design considerations included material selection (aluminum vs. copper), fin geometry, mode of convection (natural vs. forced), and fan performance. Both materials were analyzed using forced convection, as natural convection proved inadequate. Copper fins achieved higher thermal efficiency (95.29%) with 13 fins, while aluminum fins reached 88.6% efficiency with 14 fins. Despite slightly lower performance, aluminum was chosen for its cost-effectiveness and lower weight. A Noctua NF-A8 80mm fan was selected to ensure adequate airflow. The final design meets the thermal and spatial requirements, providing an efficient and economical solution for CPU cooling.

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1. INTRODUCTION

In the era of high-speed computing and miniaturized electronics, thermal management has emerged as a critical design challenge, especially in devices like central processing units (CPUs), where power densities continue to rise. Excessive heat generation within CPUs can lead to reduced performance, shortened lifespan, and in severe cases, permanent hardware damage. To mitigate these effects, thermal control systems such as heat sinks are essential components of modern electronic cooling strategies.

This project focuses on the thermal analysis and design of a heat sink intended for a CPU whose metal casing measures 80 mm × 80 mm with a thickness of 12 mm and dissipates 110 W of thermal energy under operational conditions. The CPU operates in an ambient environment of 25 °C, and the heat sink must ensure that the CPU surface temperature does not exceed 60 °C. These conditions impose strict thermal and geometric constraints, notably a maximum heat sink height of 35 mm and a minimum fin spacing of 2 mm. Therefore, the primary objective is to develop an efficient and compact heat sink capable of maintaining the CPU within safe operating temperatures while adhering to space and performance limitations.

Heat sinks function by enhancing the rate of heat transfer from a surface (in this case, the CPU casing) to the surrounding air, primarily through a combination of conduction and convection mechanisms. The design process involves careful consideration of factors such as the material selection for thermal conductivity, the number and dimensions of fins to maximize surface area, and the nature of convection—whether passive (free) or active (forced using a fan). The choice between free and forced convection has both performance and economic implications. While forced convection generally achieves higher heat transfer rates due to the increased convective heat transfer coefficient, it also requires additional power input and cost for the cooling fan, potentially affecting the system's efficiency and cost-effectiveness.

A comprehensive comparison between these two convection modes is necessary to inform the final design decision. Parameters such as the thermal resistance of the heat sink, total heat transfer rate, fin

effectiveness, and system cost will be analyzed. Material selection is also crucial; commonly used materials such as aluminum and copper offer high thermal conductivity, with aluminum often preferred due to its lightweight and cost advantages despite slightly lower conductivity compared to copper. Fin geometry also plays a significant role in determining the thermal performance, and it must be optimized within the allowable physical constraints to maximize cooling without violating design limitations.

Moreover, this project will consider economic factors in the final evaluation of the heat sink design. An ideal solution will strike a balance between thermal performance, manufacturability, operational reliability, and cost. By employing analytical modeling, thermal resistance calculations, and basic economic analysis, the design process will guide the selection of an optimal heat sink configuration tailored to the CPU's cooling needs.

Ultimately, the goal of this project is to present a complete and feasible heat sink design solution that ensures effective thermal control for the CPU while meeting spatial, thermal, and financial constraints. This task integrates foundational principles of heat transfer with practical engineering design methodology to address a real-world problem in electronic device cooling.

2. MODEL AND CALCULATIONS

DIMENSIONS OF THE FINS

$$\text{Thickness (t)} = 2\text{mm}$$

$$\text{Length (L)} = 35\text{mm}$$

$$\text{Width (W)} = 80\text{mm}$$

$$T_{\text{surr}} = 25^{\circ}\text{C}$$

$$T_{\text{surf}} = 60^{\circ}\text{C}$$

$$A_{\text{fin}} = 2 \times W \times L_c \text{ (Table 3-3)}$$

MATERIALS USED

- Aluminum 6061; $k = 152 \frac{\text{W}}{\text{mK}}$
- Copper; $k = 401 \frac{\text{W}}{\text{mK}}$

$$L_c = L + \frac{t}{2}$$

$$L_c = 0.035 + \frac{0.002}{2} = 0.036\text{m}$$

$$A_{\text{fin}} = 2 \times 0.08 \times 0.036 = 0.00576\text{m}^2$$

Choosing a flow of 70CFM

$$1 \text{ CFM} = 0.0282 \frac{\text{m}^3}{\text{min}}$$

$$V. = 0.03304 \frac{\text{m}^3}{\text{s}}$$

$$V_{\text{avg}} = \frac{V.}{A} = \frac{0.03304}{0.00576} = 5.736 \frac{\text{m}}{\text{s}}$$

$$T_{\text{film}} = \frac{T_{\text{surr}} + T_{\text{surf}}}{2}$$

$$T_{\text{film}} = \frac{60 + 25}{2} = 42.5^{\circ}\text{C}$$

Properties of air @ T_{film} (Appendix A-15)

$$\text{Pr} = 0.7248$$

$$k = 0.026605 \frac{\text{W}}{\text{m}^2\text{K}}$$

$$\nu = 1.9295 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$$

$$\text{Cp} = 1.007 \frac{\text{J}}{\text{kgK}}$$

$$\rho = 1.118 \frac{kg}{m^3}$$

Note that all properties were found using interpolation. Now, we find the Reynold's number

$$Re = \frac{V_{avg} \times L}{\nu} = \frac{5.736 \times 0.035}{1.9295 \times 10^{-5}}$$

$$Re = 10404.7 < 5 \times 10^5$$

This means that the flow is laminar

$$Nu \text{ for laminar flow} = 0.664 \times Re^{0.5} \times Pr^{0.33}$$

$$Nu = 0.664 \times 10404.7^{0.5} \times 0.7248^{0.33}$$

$$Nu = 60.91$$

$$h_{conv} = \frac{K}{L} Nu = \frac{0.026605}{0.035}$$

$$h_{conv} = 46.3 \frac{W}{m^2 K}$$

FORCED CONVECTION CALCULATIONS (Copper)

$$m = \sqrt{\frac{2h}{Kt}} = \sqrt{\frac{2 \times 46.3}{401 \times 0.002}} = 10.745 \text{m}^{-1}$$

$$mL_c = 10.745 \times 0.036 = 0.3868$$

$$\text{Fin efficiency } \eta_{\text{fin}} = \frac{\tanh(mL_c)}{mL_c}$$

$$\eta_{\text{fin}} = \frac{\tanh(0.3868)}{0.3868} = 0.9529$$

$$\eta_{\text{fin}} = 95.29\%$$

$$Q_{\text{per fin}} = \eta_{\text{fin}} \times A_{\text{fin}} \times h \times (T_{\text{surf}} - T_{\text{surr}})$$

$$Q_{\text{per fin}} = 0.9529 \times 0.00576 \times 46.3 \times (60 - 25)$$

$$Q_{\text{per fin}} = 8.894 \text{W}$$

$$\text{Number of fins } N_{\text{fin}} = \frac{Q_{\text{total}}}{Q_{\text{fin}}} = \frac{110}{8.894}$$

$$N_{\text{fin}} = 12.367$$

This means there must be at least 13 fins for the copper material to dissipate effectively

$$Q_{\text{total fin}} = 13 \times 8.894 = 115.62 \text{W}$$

$$\text{Fin effectiveness } \epsilon_{\text{fin overall}} = \frac{Q_{\text{total fin}}}{Q_{\text{total no fin}}}$$

$$\epsilon_{\text{fin overall}} = \frac{115.62}{(0.035 \times 0.08) \times 46.3 \times (60 - 25)}$$

$$\epsilon_{\text{fin overall}} = 25.48$$

FORCED CONVECTION SIMULATION (Copper)

Table 2-1 Simulation Input for forced convection of copper

material	Copper
Flow Type	Laminar and Turbulent
Gravity	9.81m/s
Fluid	Air
Analysis Type	External
Ambient temp	25
Pressure	101325 pa
power	110W
Velocity	5.5m/s

In this setup, copper is subjected to the same forced convection environment as aluminum. The outcome highlights copper's exceptional thermal management capabilities: **it achieves the lowest maximum temperature of all cases studied.** The forced airflow rapidly removes heat conducted to the surface, taking full advantage of copper's high thermal conductivity. This combination leads to extremely effective cooling, ideal for applications where thermal performance is critical—such as in electronics, high-power components, and precision instruments. The simulation illustrates how copper, although more expensive, excels in both passive and active cooling systems, especially when high reliability is needed.

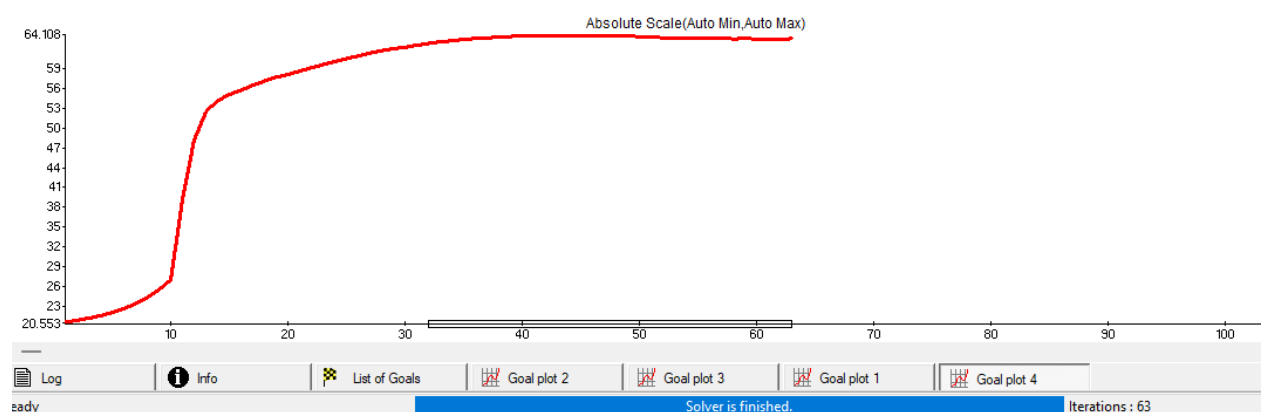


Figure 2.1 Max temperature for forced convection

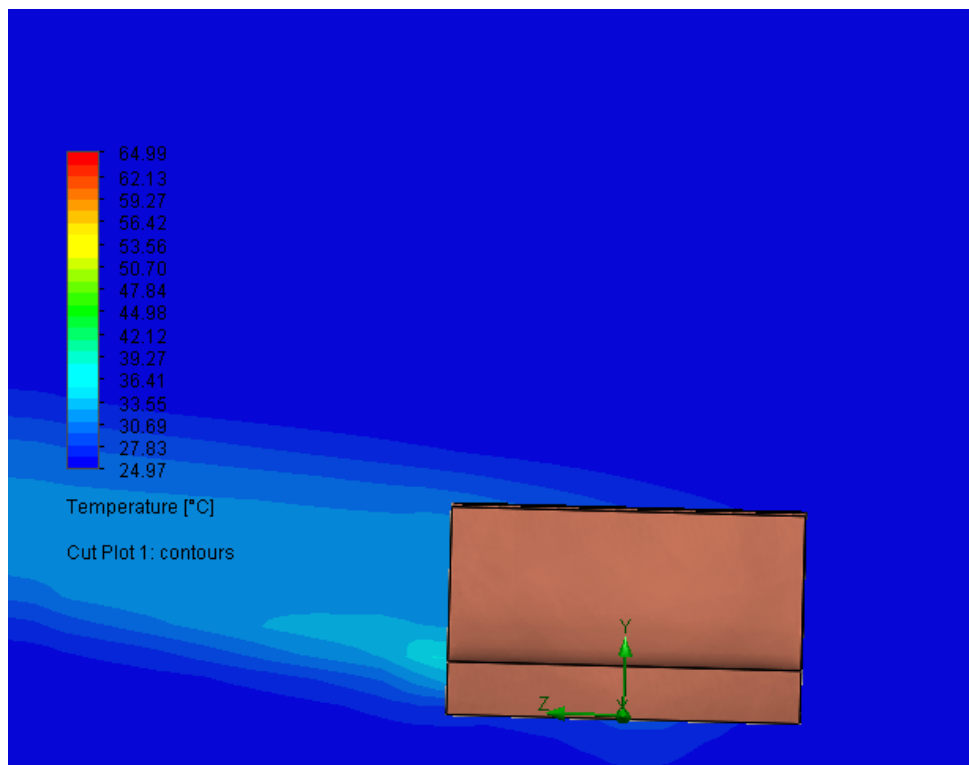


Figure 2.2 Contours for forced convection

FORCED CONVECTION CALCULATIONS (Aluminum)

$$m = \sqrt{\frac{2h}{Kt}} = \sqrt{\frac{2 \times 46.3}{152 \times 0.002}} = 17.45\text{m}^{-1}$$

$$mL_c = 17.45 \times 0.036 = 0.6283$$

$$\text{Fin efficiency } \eta_{\text{fin}} = \frac{\tanh(mL_c)}{mL_c}$$

$$\eta_{\text{fin}} = \frac{\tanh(0.6283)}{0.6283} = 0.886$$

$$\eta_{\text{fin}} = 88.6\%$$

$$Q_{\text{per fin}} = \eta_{\text{fin}} \times A_{\text{fin}} \times h \times (T_{\text{surf}} - T_{\text{surr}})$$

$$Q_{\text{per fin}} = 0.886 \times 0.00576 \times 46.3 \times (60 - 25)$$

$$Q_{\text{per fin}} = 8.273\text{W}$$

$$\text{Number of fins } N_{\text{fin}} = \frac{Q_{\text{total}}}{Q_{\text{fin}}} = \frac{110}{8.273}$$

$$N_{\text{fin}} = 13.29$$

This means there must be at least 14 fins for the copper material to dissipate effectively

$$Q_{\text{total fin}} = 14 \times 8.273 = 115.82\text{W}$$

$$\text{Fin effectiveness } \epsilon_{\text{fin overall}} = \frac{Q_{\text{total fin}}}{Q_{\text{total no fin}}}$$

$$\epsilon_{\text{fin overall}} = \frac{115.82}{(0.035 \times 0.08) \times 46.3 \times (60 - 25)}$$

$$\epsilon_{\text{fin overall}} = 25.525$$

FORCED CONVECTION SIMULATION (Aluminum)

Table 2-2 Simulation input for forced convection on Aluminum fin

material	Aluminum 6061
Flow Type	Laminar and Turbulent
Gravity	9.81m/s
Fluid	Air
Analysis Type	External
Ambient temp	25
Pressure	101325 pa
power	110W
Velocity	5.5m/s

This simulation evaluates aluminum's performance when exposed to forced convection using a 5.5 m/s airflow. The presence of external airflow substantially enhances convective heat transfer by thinning the boundary layer and removing heat from the surface more effectively. The result is a

marked reduction in surface and maximum temperatures compared to the natural convection scenario. Aluminum's relatively lower cost and lighter weight make it a popular material in cooling applications, and when paired with forced airflow, it becomes a viable solution for dissipating moderate heat loads efficiently

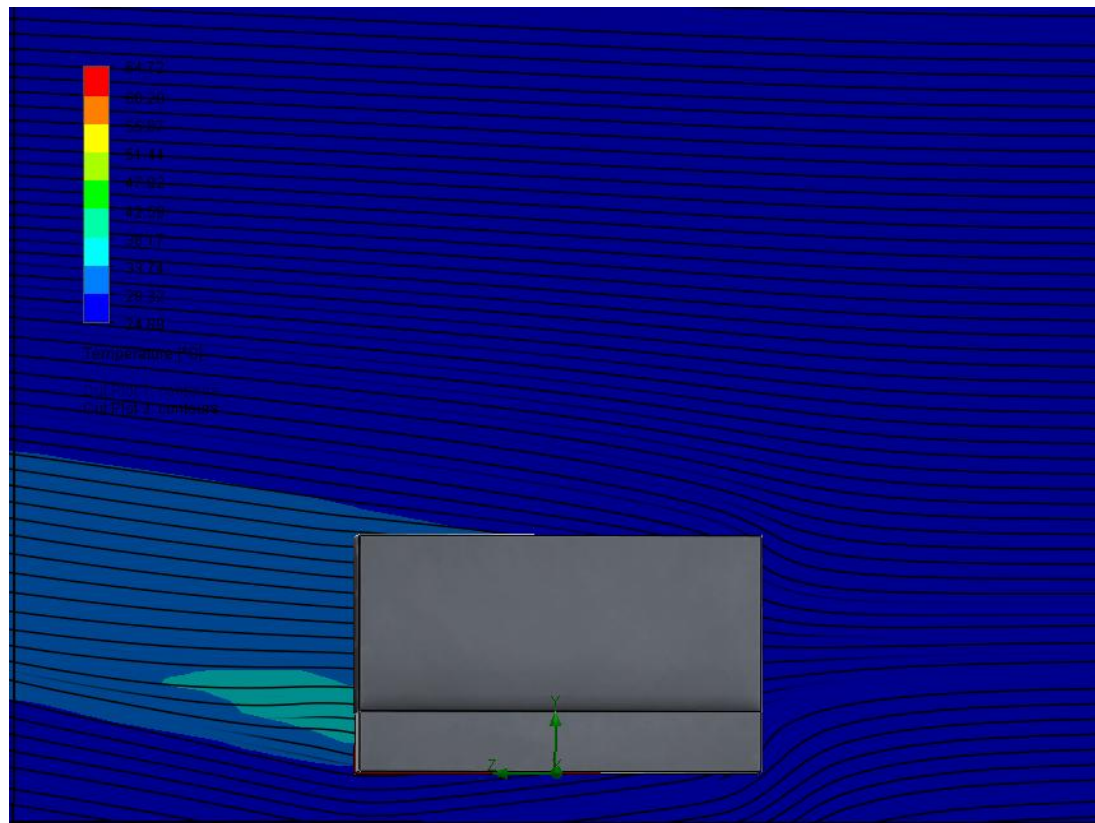


Figure 2.3 Contours for forced convection on Aluminum Fin

In summary, copper is more effective to use with 14 fins and an efficiency of 95% but it is more costly than aluminum which is cost friendly and has an efficiency of 88% while using 13 fins.

When selecting a fan, we need the following properties

- Velocity – above 5.5 m/s
- Dimensions – 80mm x 80mm
- CFM – 70CFM

From the given assumption and calculations aluminum fin is preferable to use

NATURAL CONVECTION

$$T_{\text{film}} = 42.5^{\circ}\text{C (unchanged)}$$

$$\nu = 1.9295 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7248, K = 0.026605 \text{ W/m.k}$$

$$\beta = \text{volume expansion} = 1/315.65 = 3.168 \times 10^{-3}$$

$$\text{Ra}_D = \frac{g\beta(T_s - T_{\text{surr}})L_c^3}{\nu^3} \times \text{Pr}$$

Where:

$$\beta = 0.003168, T_s = 60^{\circ}\text{C}, g = 9.81 \text{ m/s}^2, T_{\text{surr}} = 25^{\circ}\text{C}, L_c = A_s/P \text{ and } \nu = 1.9295 \times 10^{-5} \text{ m}^2/\text{s}$$

$$L_c = (0.08 \times 0.035) / (2 \times (0.08 + 0.035)) = 0.01217$$

$$\text{Ra}_D = \frac{9.81 \times 0.003168 (60 - 25) 0.01217^3}{(1.9295 \times 10^{-5})^3} \times 0.7248$$

Nusselts number for finned surfaces:

$$\text{Nu} = h_L S / K \text{ where } S = S_{\text{opt}}$$

$$S_{\text{opt}} = 2.714 L / \text{Ra}_D^{0.25} = 0.035 / 5266.28^{0.25} \times 2.714 = 0.01115 \text{ m}$$

$$S_{\text{opt}} = 0.01115 \text{ m}$$

$$\text{Nu} = h S_{\text{opt}} / K = 1.307$$

$$h = (1.307 \times K) / S_{\text{opt}} = (1.307 \times 0.026605) / 0.01115 = 3.12 \text{ W/m.k}$$

$$h = 3.12 \text{ W/m.k}$$

to find rate of heat transfer by natural convection from fins

$$Q = h \times (2nLH) \times (T_s - T_{\text{surr}})$$

$$N = W / (S + t) = 0.08 / (0.01115 + 0.002) = 6.0836 \text{ (number of fins at least 6)}$$

$$Q = 3.12 (2 \times 6.0836 \times 0.035 \times 0.08) \times (60 - 25) = 3.72 \text{ W}$$

$$\text{Gr} = \text{Re} / \text{Pr} = 10404.278 / 0.7248 = 14355.27$$

Checking, $\text{Gr} / \text{Re}^2 = 14355.27 / 10404.7^2 = 0.0001326$, which is $\ll 1$. Therefore, we used forced convection instead.

NATURAL CONVECTION SIMULATION (Aluminum)

Table 2-3 Simulation input for Natural Convection on Aluminum Fin

material	Aluminum 6061
Flow Type	Laminar and Turbulent
Gravity	9.81m/s
Fluid	Air
Analysis Type	External
Ambient temp	25
Pressure	101325 pa
power	110W
Velocity	0m/s

This simulation models the thermal response of Aluminum 6061 under natural convection conditions, where air movement is driven solely by buoyancy effects resulting from temperature differences between the heated surface and the ambient air. The results show that aluminum, while possessing moderate thermal conductivity, experiences **higher surface and maximum temperatures** than copper under identical settings. The absence of forced airflow limits the rate of heat dissipation, leading to slower cooling and a thicker thermal boundary layer. This highlights aluminum's limitations when used in passively cooled systems without any mechanical assistance.

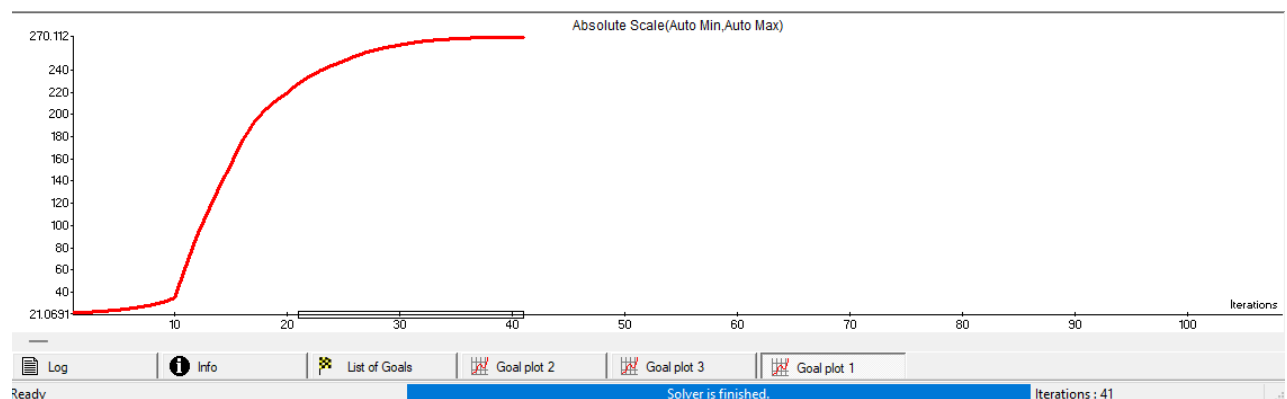


Figure 2.4 Max Temperature for Natural Convection on Aluminum fin

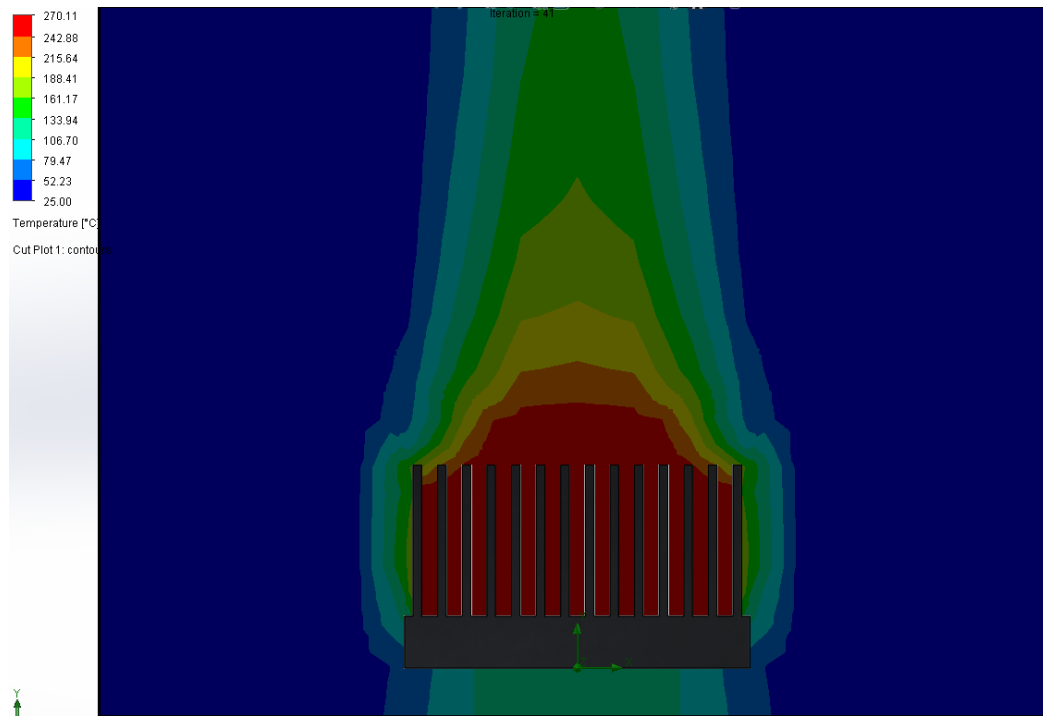


Figure 2.5 Contours Natural Convection for Aluminum

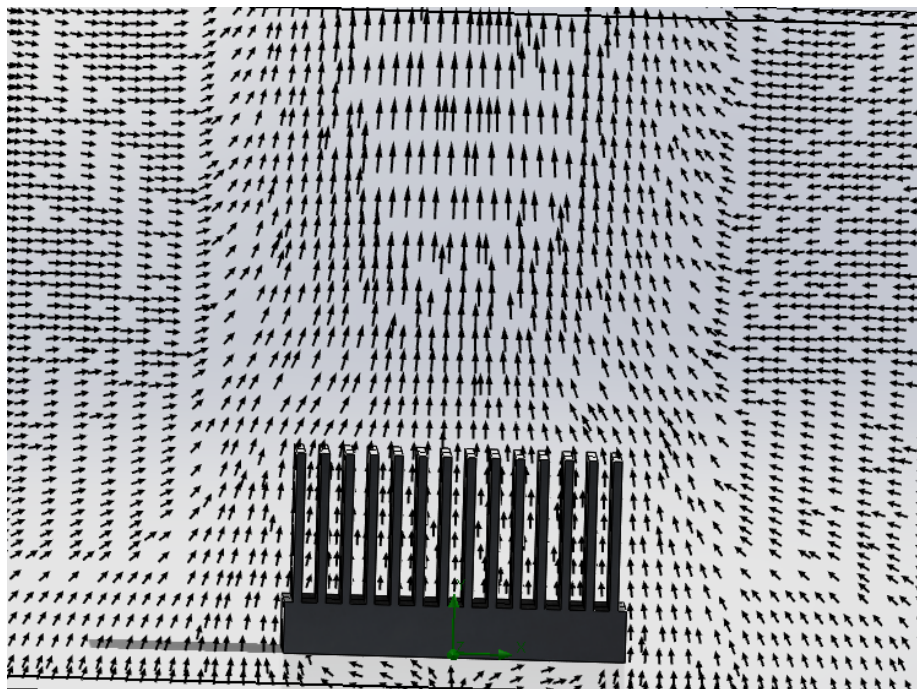


Figure 2.6 Natural Convection Buoyancy Effect

NATURAL CONVECTION SIMULATION (copper)

Table 2-4 Simulation input for Natural Convection of Copper

material	copper
Flow Type	Laminar and Turbulent
Gravity	9.81m/s
Fluid	Air
Analysis Type	External
Ambient temp	25
Pressure	101325 pa
power	110W
Velocity	0m/s

In the copper-based natural convection simulation, the system demonstrates noticeably improved thermal performance compared to aluminum. Due to copper's superior thermal conductivity (which is more than twice that of aluminum), heat is conducted away from the source more rapidly and transferred to the surrounding air more efficiently. As a result, **the maximum temperatures observed are significantly lower** than those seen in the aluminum case. Although both materials rely on buoyancy-driven flow, copper enables thinner thermal boundary layers and more uniform heat distribution, making it a better choice for passive cooling applications.

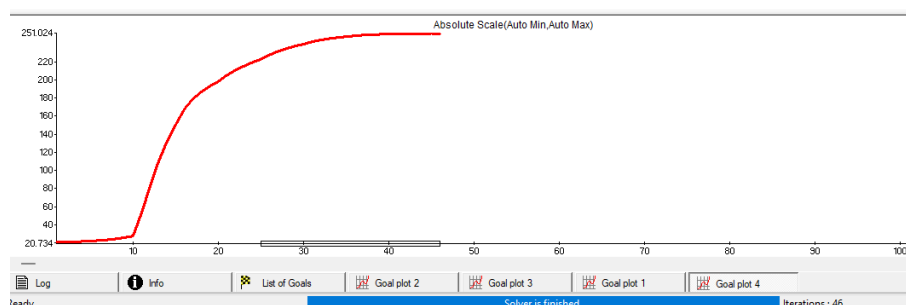


Figure 2.7 Max Temperature for Natural Convection of Copper

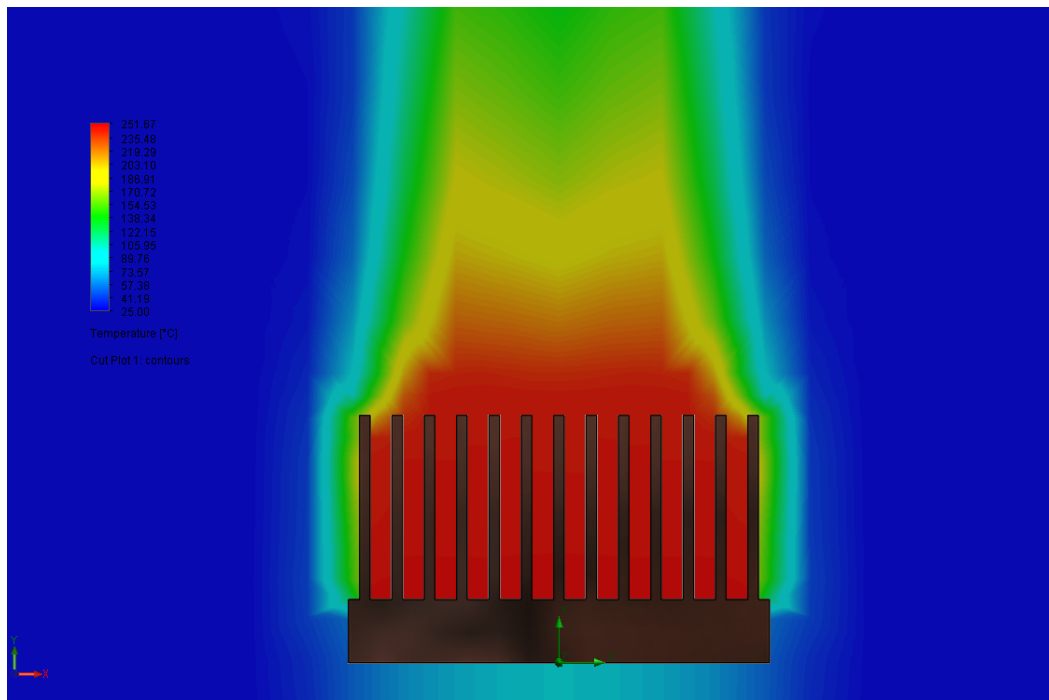


Figure 2.8 Contours for Copper Fin

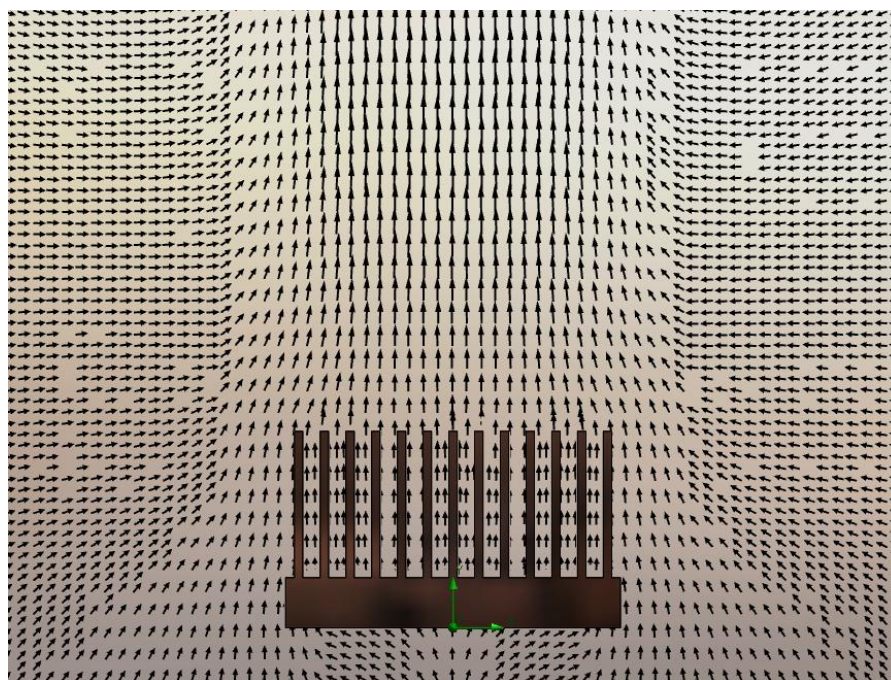


Figure 2.9 Buoyancy Effect on Copper Fin

DISCUSSION OF SIMULATION RESULTS

Based on the simulation results, it is evident that both the type of convection and the thermal conductivity of the material significantly influence heat dissipation performance. Under natural convection, copper outperforms aluminum by achieving lower maximum temperatures due to its higher thermal conductivity, making it more effective in passive cooling scenarios. However, both materials exhibit limited cooling capacity without airflow. When forced convection is introduced, there is a substantial improvement in heat transfer for both materials, with copper again showing the best performance by reaching the lowest overall temperatures. Therefore, while copper is the superior material in all cases, the use of **forced convection** is strongly recommended for applications with higher heat loads or where thermal regulation is critical, as it drastically enhances cooling efficiency regardless of material choice.

3. FAN SELECTION

- Cooler Master i7C RGB (intel only)



Figure 3.1 - Cooler Master i7C RGB Fan

Table 3-1 - Cooler Master i7C RGB specifications

Product dimension	4.72"L x 2.36"W x 4.72"H
Brand	Cooler master
Power Connector Type	4 Pin type
voltage	12 Volts
wattage	2.4
price	23 USD

- Noctua NF-A8 80mm pwm fan



Figure 3.2 - Noctua NF-A8 80mm pwm fan

Table 3-2 - Noctua NF-A8 80mm pwm fan Specifications

Product dimension	3.15"L x 0.98"W x 3.15"H
Brand	Noctua
Power Connector Type	4 Pin type
voltage	12 Volts
wattage	2.4
price	17 USD

Two fans were evaluated for forced convection: the **Cooler Master i7C RGB**, and the **Noctua NF-A8 80mm PWM**. Both operated at 12V and consumed 2.4W, but the Noctua fan offered a more compact design and a better price-to-performance ratio at **\$17**, compared to Cooler Master's **\$23**. Based on thermal and economic analysis, the Noctua fan was selected.

Therefore, the forced convection aluminum heat sink with 14 fins and the Noctua fan offers a highly efficient and economically viable cooling solution. It meets all spatial and thermal constraints while maintaining operational reliability and cost-effectiveness.

4. RESULTS AND DISCUSSION

The final design results provide a comparative analysis of two potential materials; copper and aluminum, for the fabrication of the heat sink, evaluated under both forced and natural convection conditions. The goal was to ensure the CPU's surface temperature remains below 60 °C while dissipating 110 W of thermal energy, constrained by the maximum dimensions of 80 mm × 80 mm base area and 35 mm height.

Forced Convection Analysis:

Under forced convection with a fan delivering 70 CFM (velocity ≈ 5.736 m/s), the copper heat sink achieved a **fin efficiency of 95.29%**, with each fin dissipating approximately **8.89 W**. To meet the thermal load, at least **13 fins** were required, resulting in a total heat dissipation of **115.62 W**, surpassing the necessary 110 W. The **overall fin effectiveness** was calculated as **25.48**, indicating excellent performance in enhancing heat transfer.

Aluminum, in comparison, achieved a slightly lower **efficiency of 88.6%**, with **each fin dissipating 8.27 W**. However, due to the reduced thermal conductivity of aluminum (152 W/m·K compared to copper's 401 W/m·K), **14 fins** were needed to meet the thermal requirement. The total heat dissipation was calculated as **115.82 W**, with a fin effectiveness of **25.52**, which is comparable to copper despite the lower material performance.

While copper performed slightly better thermally, the difference was marginal when considering the total heat transfer. The decision ultimately leaned toward **aluminum** due to

cost-effectiveness, availability, and lower weight—critical factors in large-scale manufacturing and CPU cooling systems.

Natural Convection Analysis:

When evaluated under natural convection, the calculated **heat transfer coefficient was only 3.12 W/m²·K**, resulting in a total heat dissipation of just **3.72 W** from six fins—far below the 110 W requirement. Additionally, the dimensionless Gr/Re² ratio was significantly less than 1 (≈ 0.00013), confirming the dominance of forced convection for the application. Natural convection alone proved insufficient for the heat sink's intended purpose, making forced convection a necessity.

5. CONCLUSION

The heat sink design project successfully addressed the thermal management challenges posed by a CPU dissipating 110 W within a constrained volume. Through detailed thermal analysis and comparison of materials, it was determined that **forced convection** is the only viable cooling method for this application, as **natural convection** yielded insufficient heat dissipation (only 3.72 W).

Between the two materials analyzed the **Copper** and **aluminum**. The Copper demonstrated superior thermal performance, with a fin efficiency of 95.29% and the ability to dissipate 115.62 W using 13 fins. However, aluminum offered comparable performance (88.6% efficiency and 115.82 W dissipation with 14 fins) at a lower cost and weight. Given the marginal difference in thermal output and the advantages in manufacturability and affordability, **aluminum was selected as the optimal material**.

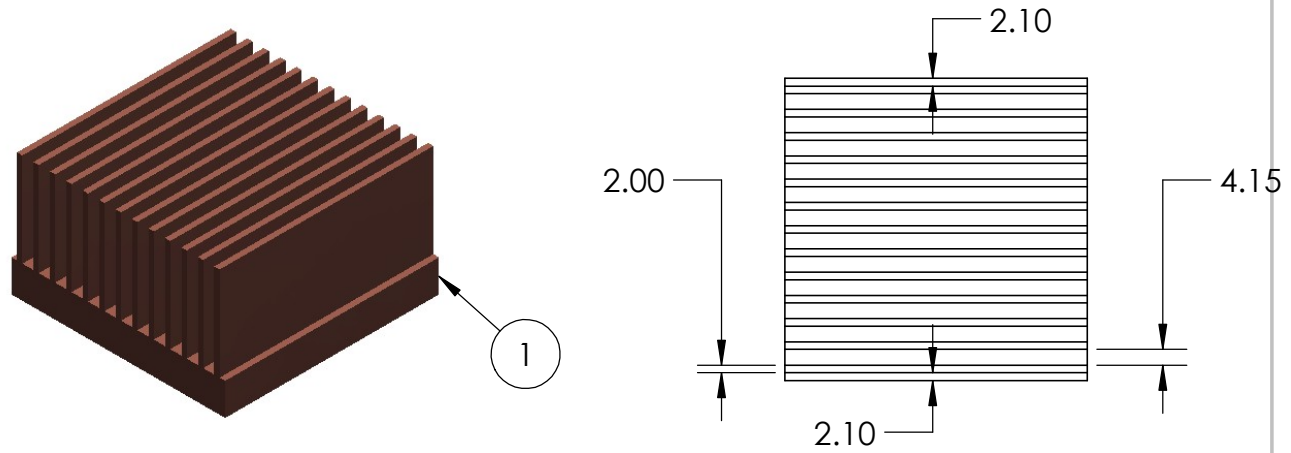
A **Noctua NF-A8 80mm PWM fan** was chosen for forced convection due to its suitable airflow characteristics (≥ 70 CFM), compact dimensions, and economic value. Combined with the aluminum fins, this fan ensures the heat sink maintains the CPU temperature below the critical limit of 60°C during operation.

Overall, the final heat sink design—comprising **14 aluminum fins** and an **efficient forced-air fan**—provides a robust, compact, and cost-effective solution. The design satisfies all spatial, thermal, and economic constraints, demonstrating the effective application of heat transfer principles to a real-world engineering problem

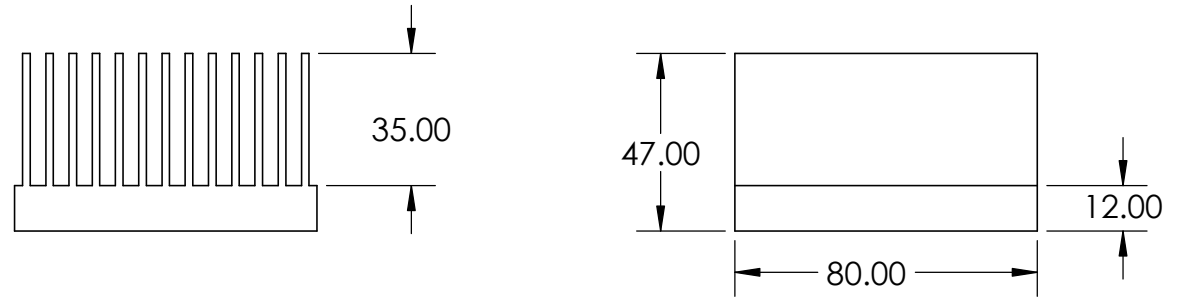
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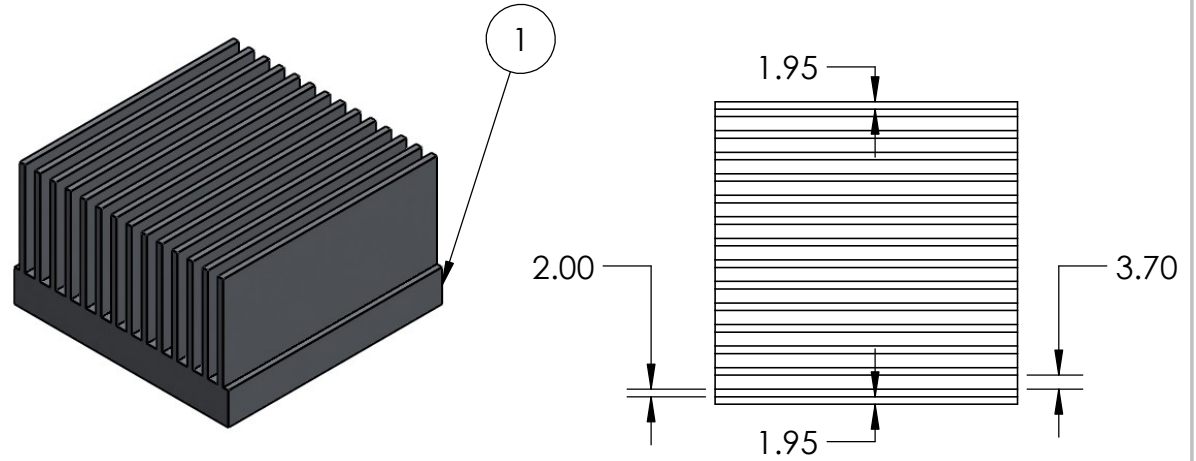
7. APPENDIX



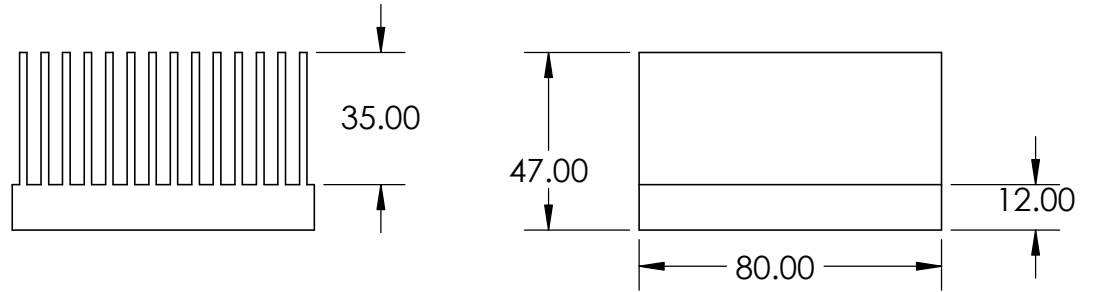
ITEM NO.	MATERIAL	QTY.
1	Copper	1



	NAME:	DATE:	SIGNATURE:	EMU
DRAWN BY:	Tarik. T	5/27/25	T.T	
CHECKED BY:	Professor Murat			
Units: mm Scale:1:2	COPPER FIN			DRAWING NUMBER: 1-1-1



ITEM NO.	MATERIAL	QTY.
1	6061 Alloy	1



	NAME:	DATE:	SIGNATURE:	EMU
DRAWN BY:	Tarik. T	5/27/25	T.T	
CHECKED BY:	Professor Murat			
Units: mm Scale: 1:2	ALUMINIUM FIN			DRAWING NUMBER: 1-1-1